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Physics-Informed Broad Learning System for Lithium-Ion Battery Prognostics: A Shallow Architecture Approach

Weiheng Li
Jinan University & University of Birmingham Joint Institute, Jinan University
Guangzhou 511443, China
liweuheng2023101693@stu2023.jnu.edu.cn

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***Abstract*—Physics-informed neural networks promise to reconcile physical consistency with data-driven flexibility for battery prognostics, yet suffer from gradient instability when backpropagating through deep architectures under stiff differential equation constraints. We demonstrate that shallow, wide networks achieve superior performance while sidestepping this fundamental limitation. By embedding Verhulst degradation models and DeepHPM-discovered dynamics into Broad Learning Systems—computing gradients through at most two layers—we preserve well-conditioned optimization landscapes that deep physics-informed networks cannot maintain. Validated on 124 LFP/graphite cells across distribution shifts (cross-protocol, mixed-protocol scenarios), our Physics-Informed Broad Learning System (PIBLS) reduces RUL prediction error by 48–61% compared to deep baselines while training 10–100× faster. A multi-activation ensemble (identity, tanh, ReLU, sine) maps distinct timescales onto physical processes (drift, smoothness, sparsity, periodicity), contributing 37% improvement over single-activation designs. Beyond battery applications, this work suggests a broader lesson: when physical consistency matters, architectural simplicity often outperforms complexity. Shallow topologies preserve gradient stability under physics-based regularization—an advantage critical for scientific machine learning in stiff, multi-scale systems.**

***Index Terms*—Physics-Informed Neural Networks, Broad Learning Systems, Battery Prognostics, Multi-Scale Dynamics, Gradient Stability**

I. Introduction

A battery management system predicts 500 cycles remaining; the cell fails after 50. This is not a hypothetical scenario—such catastrophic mispredictions occur regularly in deployed electric vehicle fleets, where data-driven models trained on laboratory conditions encounter distribution shifts from real-world temperature swings, aggressive driving, and fast-charging abuse [1]. The stakes are high: with global lithium-ion battery markets projected to exceed $400 billion by 2030 [2], [3], even small improvements in prognostic accuracy translate to billions in avoided warranty costs and enhanced safety. Yet current approaches remain trapped in a paradox: physics-based models promise reliability but demand intractable calibration; pure machine learning achieves flexibility but produces physically nonsensical predictions (capacity increases, negative resistance) that collapse outside training regimes. This tension between physical rigor and data-driven adaptability has stalled progress in battery health prediction for over a decade.

Battery prognostics centers on two critical tasks: estimating State of Health (SOH)—the ratio of current to nominal capacity—and predicting Remaining Useful Life (RUL)—the cycles until reaching failure threshold (typically 80% capacity retention). These tasks are fundamentally challenging for three reasons. First, battery degradation exhibits highly nonlinear dynamics driven by coupled electrochemical processes including solid-electrolyte interphase (SEI) growth, active material loss, and lithium inventory depletion [4]–[6]. Second, individual cells manifest distinct aging trajectories due to manufacturing variance and usage history, demanding models that generalize across populations while respecting physical constraints [1]. Third, real-world operating conditions—varying charge protocols, temperature profiles, and load patterns—introduce distribution shifts that expose the brittleness of purely empirical models [7], [8].

Existing approaches fall into three camps, each with fundamental limitations rooted in architectural mismatches. Physics-based models (e.g., Doyle-Fuller-Newman [9]) embed electrochemical equations but require solving stiff partial differential equations with numerous parameters that are expensive to calibrate and non-identifiable from degradation curves alone [10]. This stems from a core issue: first-principles models describe microscopic mechanisms (ion diffusion, reaction kinetics) while measurements reflect macroscopic outcomes (capacity fade), creating an inverse problem that is ill-posed without extensive characterization data. Empirical models (e.g., polynomial or exponential fits [11]) sidestep physics entirely, achieving computational efficiency at the cost of extrapolation ability—they interpolate well within training regimes but diverge unpredictably beyond. Data-driven methods, particularly deep neural networks [12], offer flexibility but ignore conservation laws and thermodynamic constraints, occasionally predicting capacity increases or negative resistance values [13].

Recent physics-informed neural network (PINN) approaches [14] attempt to bridge this gap by regularizing deep networks

Corresponding author. Email: liweuheng2023101693@stu2023.jnu.edu.cn

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with differential equation residuals. However, they inherit a critical design flaw: backpropagating through multiple stacked layers to satisfy physical constraints amplifies gradient instability, especially for the stiff, multi-scale dynamics characteristic of battery degradation. The deeper the network, the harder it becomes to balance data fidelity against physical consistency—a phenomenon manifested in the notorious difficulty of tuning loss weights [15]. This reveals a fundamental tension: physics requires smooth, well-conditioned function approximators to compute accurate derivatives, yet deep architectures introduce conditioning challenges that worsen with depth. Could we reconcile physical rigor with training stability by rethinking the network topology itself?

Research Gap and Motivation. While physics-informed approaches show promise, they rest on an unexamined assumption that has limited progress: that *depth is necessary for modeling complex physical systems*. We challenge this assumption by posing a fundamental question: Can shallow architectures achieve superior physics-informed learning by preserving gradient conditioning? This question is motivated by three critical observations from deployment realities. First, battery degradation PDEs exhibit condition numbers exceeding 10^6 in multi-scale regimes (millisecond transients to month-long drift), causing gradient flow through deep networks to become pathologically unstable—training diverges or converges to physically inconsistent solutions. Second, real-world battery management systems demand sub-second inference and require models trainable within hours, yet current PINNs need 10–100$\times$ more training time than pure data-driven methods, creating an adoption barrier. Third, existing deep PINNs exhibit over 50% error sensitivity to loss weight configurations across $\lambda \in [0.5, 2.0]$ and 18.7% variance across random initializations, rendering deployment unreliable without extensive hyperparameter search. These challenges—gradient pathology, computational cost, and optimization brittleness—collectively demand a fundamental rethinking of network topology for scientific machine learning. Rather than incrementally improving deep architectures, we ask whether architectural simplicity itself can be a solution.

We show that a century-old insight from machine learning—that width, not depth, can achieve universal approximation—offers a surprisingly effective resolution when paired with automatic differentiation. Our key observation is that Broad Learning Systems (BLS) [16], which employ shallow, wide architectures with random feature mappings, provide naturally well-conditioned gradients for physics-based regularization. Unlike deep networks where gradients traverse many nonlinear transformations, BLS computes derivatives through at most two layers, dramatically stabilizing the joint optimization of data loss and physical residuals. This echoes classical numerical analysis wisdom: solving differential equations benefits from good problem conditioning, which shallow architectures preserve by construction.

Building on this principle, we introduce Physics-Informed Broad Learning System (PIBLS), a framework that embeds physical laws into BLS through three synergistic components. First, for SOH estimation, we incorporate the Verhulst differential equation [17], [18]—a logistic growth model capturing the S-shaped degradation trajectory from initial SEI formation to saturation—directly into the loss function via automatic differentiation. Second, for RUL prediction lacking explicit governing equations, we employ DeepHPM [19] to discover latent dynamics from data while constraining solutions to obey learned partial differential equations. Third, recognizing that battery degradation involves multiple timescales (fast voltage transients, slow capacity fade), we design a multi-activation strategy combining identity (linear trends), tanh (smooth nonlinearity for differentiability), ReLU (sparse feature extraction), and sinusoidal (periodic effects) functions distributed across feature nodes. These activations address the challenge of representing multi-scale physics: linear components capture long-term drift, periodic terms model cyclic stress, and nonlinear units fit transient anomalies. To address the multi-objective optimization challenge, we adopt uncertainty-weighted adaptive loss balancing [20], treating weights as learnable parameters that automatically adjust the contribution of data fidelity, PDE residuals, and temporal smoothness constraints during training.

We validate PIBLS on the A123 LFP/graphite battery dataset comprising 124 cells cycled under diverse fast-charging protocols until failure. Across three experimental scenarios—same-protocol transfer (Case A), different-protocol generalization (Case B), and mixed-protocol deployment (Case C)—PIBLS achieves substantial improvements over baselines. For RUL prediction, PIBLS reduces RMSE by 48.0% compared to Bayesian-LSTM (38.36 vs. 19.96 cycles in Case A), 23.9% versus standard PINN (37.90 vs. 30.59 cycles in Case B), and 60.6% versus deep PINN (32.50 vs. 12.81 cycles in Case C). For SOH estimation, PIBLS achieves 24.8% lower RMSPE than physics-informed deep networks (0.00290 vs. 0.00218 in Case B). Critically, these gains persist in cross-protocol scenarios where training and test cells follow different charging schedules—the distribution shifts that expose brittleness in purely data-driven models. Ablation studies confirm that multi-activation strategies contribute 36.6% error reduction over single-activation BLS (31.50 vs. 19.96 RMSE in Case A), and adaptive weighting improves stability in challenging generalization settings (Case B).

The main contributions of this work are:

- **Architectural innovation**: We demonstrate that shallow, wide networks (BLS) combined with physics-based regularization outperform deep physics-informed networks for battery prognostics, achieving 48–61% error reduction on RUL tasks while maintaining gradient stability—challenging the prevailing assumption that depth is necessary for complex physical systems.
- **Multi-scale representation learning**: We introduce a principled multi-activation ensemble (identity, tanh, ReLU, sine) that maps onto distinct degradation mechanisms (long-term drift, smooth nonlinearity, sparse events, cyclical stress), verified through ablations showing 37% improvement over single-activation designs.

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- **Empirical validation**: We provide the first systematic comparison of BLS-based physics-informed learning across three transfer scenarios (same-protocol, cross-protocol, mixed-protocol) on a large-scale battery dataset, demonstrating consistent 24–61% gains over PINN, Bayesian-LSTM, and Gaussian process baselines with statistical significance.

Beyond battery applications, these findings suggest a broader lesson for scientific machine learning: when physical consistency matters, architectural simplicity (shallow+wide) may outperform architectural complexity (deep+narrow), especially under distribution shift. The approach assumes degradation follows smooth differential dynamics; abrupt failure modes (thermal runaway, internal shorts) remain outside scope. The remainder of this paper is organized as follows: Section 2 reviews related work on battery prognostics and physics-informed learning; Section 3 details the PIBLS methodology; Section 4 presents experimental setup and datasets; Section 5 analyzes results and ablations; Section 6 concludes with broader implications and future directions.

II. Related Work

Battery health prognostics has evolved through three distinct paradigms over the past decade, each addressing limitations of its predecessor while introducing new challenges (summarized in Table I). Early work (2010–2015) focused on physics-based electrochemical models rooted in first-principles PDEs, achieving interpretability at the cost of parameter identifiability [9]. The deep learning era (2016–2020) introduced data-driven methods that achieved flexibility but violated physical laws under distribution shifts [12]. Recent hybrid approaches (2021–present) attempt to combine both paradigms via physics-informed neural networks, yet struggle with gradient instability in deep architectures [21]. We organize our review around the fundamental trade-off between physical rigor (interpretability, extrapolation) and computational tractability (parameter identifiability, gradient stability, training efficiency), positioning our work at the intersection where architectural simplicity enables both. Table I provides a comparative overview, highlighting how our approach addresses limitations across all three families.

Battery health prognostics—focused on State of Health (SOH) estimation and Remaining Useful Life (RUL) prediction—has evolved along three mechanism-driven paths: (A) *Physics-based electrochemical models* solving PDEs for microscopic ion transport [9]; (B) *Data-driven machine learning methods* fitting patterns from cycling data without explicit physics [12]; and (C) *Physics-informed hybrid approaches* embedding differential equation constraints into neural networks [21]. This taxonomy reveals a fundamental trade-off: physical rigor (interpretability, extrapolation) versus computational tractability (parameter identifiability, gradient stability).

A. Physics-Based Electrochemical Models

These methods numerically solve coupled PDEs governing lithium transport, electrolyte diffusion, and charge transfer [9]. The Doyle-Fuller-Newman (DFN) model requires 20+ parameters (diffusion coefficients, reaction rates) typically obtained from destructive characterization [10], solving stiff algebraic-differential systems via ODE solvers then mapping voltage curves to capacity fade. Recent neural surrogates [25] enable ¡1s predictions with ¡5% SOH error, but cross-cell generalization remains poor: models calibrated on one batch exhibit ¿15% RUL error on different manufacturers [6]. Physics-based models excel within calibration ranges and provide mechanistic interpretability [9].

The fundamental limitation is an *ill-posed inverse problem*: PDEs describe microscopic physics while measurements capture macroscopic outcomes, rendering 20+ parameters non-identifiable [10]. Optimization landscapes exhibit flat valleys—different parameter sets produce identical voltage predictions but diverge catastrophically for RUL extrapolation. Performance degrades under manufacturing variance (±5% electrode thickness causes ±20% RUL error [6]), novel charge protocols [7], and computational budgets (DFN needs seconds per cycle versus ¡100ms BMS requirements [4]). The root cause is structural rigidity: PDEs encode physics as rigid equations instead of learnable constraints, motivating methods that embed physics as *soft constraints* preserving adaptability.

B. Data-Driven Machine Learning Methods

These methods learn mappings $f_\theta : \mathbf{X} \to y$ from health indicators to degradation outcomes without explicit physics. Approaches range from Gaussian process regression [23] to deep architectures (LSTM [22], Transformers [24]) that capture temporal dependencies via universal approximation [12]. Representative results: GPR achieved 9% RUL error on 124 LFP cells but degraded to 25% under cross-protocol transfer [23]; deep LSTM reduced error to 12% on NASA datasets but required 200+ cells and exhibited 30% error across chemistries [22]. Data-driven methods excel with abundant in-distribution data, achieving millisecond inference suitable for embedded deployment and automatic discovery of nonlinear feature interactions [12].

The fundamental limitation is *physics-agnostic function fitting*: models minimize empirical risk without enforcing thermodynamic constraints [13]. Neural networks interpolate smoothly within training distributions but exhibit arbitrary behavior outside—an LSTM trained at 25°C may predict capacity *increases* at 40°C (physically impossible). Performance collapses under distribution shifts (40% error spikes on dynamic profiles [8]), sparse early predictions (GPR variance explodes with ¡50 cycles [26]), and cross-chemistry transfer (LFP models fail on NMC [1]). The root cause is *missing physical constraints*: standard loss functions cannot distinguish physically plausible from implausible extrapolations, risking safety-critical failures. This motivates architectures where physics and data *jointly* define the hypothesis class.

C. Physics-Informed Hybrid Approaches

These methods embed differential equation residuals as regularization: PINNs minimize $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{PDE}}$, where

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TABLE I
TAXONOMY OF BATTERY PROGNOSTICS APPROACHES

Category	Representative Works	Key Strength	Critical Limitation
Physics-based	DFN [9], P2D [10]	Mechanistic interpretability	Parameter non-identifiability
Data-driven	LSTM [22], GPR [23], Transformer [24]	Flexible pattern learning	Physics violations under shift
Hybrid (Deep PINN)	Verhulst-PINN [18], DeepHPM [14]	Balances physics+data	Gradient instability
PIBLS (Ours)	**Shallow BLS+Physics**	**Stable gradients, 10–100× faster**	**Assumes smooth dynamics**

$\mathcal{L}_{\text{PDE}}$ penalizes violations of governing equations via automatic differentiation [21]. Variants use explicit PDEs (Verhulst [18]) or learned dynamics (DeepHPM [19]). Representative results: PINNs with Verhulst achieved 8% SOH error versus 12% for LSTM but required careful loss weight tuning and exhibited 18% error on cross-protocol transfer [18]; DeepHPM reduced RUL error to 10% on same-protocol tests but degraded to 22% across chemistries [14]. Hybrid methods enforce conservation laws, preventing thermodynamic violations [27], and improve sample efficiency (20–30 cells versus 100+ for pure ML [18]).

The fundamental limitation is *gradient instability in deep architectures under stiff PDE constraints*. Satisfying differential equations requires accurate higher-order derivatives $\nabla^2 u_\theta$ backpropagated through multiple layers. For stiff dynamics (battery degradation spans milliseconds to months), gradients vanish or explode, causing training divergence [15], [28]. Depth compounds ill-conditioning: Hessian condition numbers exceed 10^6 in multi-scale physics [14], turning physics constraints into optimization barriers. Performance degrades via loss weight conflicts (increasing λ collapses data fit; decreasing λ loses physics [15]), initialization sensitivity (random seeds alter error by 50% [14]), and computational cost (10–100× more epochs than pure ML [18]). The root cause is *depth-physics incompatibility*: smooth differential operators clash with many-layered nonlinearities. This motivates *shallow* architectures preserving gradient well-conditioning by construction.

D. Positioning

Our approach is closest to **Family C** (physics-informed hybrids) but differs fundamentally in *network topology*: we replace deep architectures with Broad Learning Systems (BLS)—shallow, wide networks computing gradients through at most two layers. This addresses the gradient instability root cause (Introduction Gap L1) by eliminating depth-induced conditioning pathologies while maintaining physics-based regularization. We employ multi-activation ensembles (identity, tanh, ReLU, sine) to capture multi-scale degradation dynamics, bridging the expressivity-stability trade-off that plagues deep PINNs. This architectural choice—width over depth for physics constraints—remains underexplored in battery prognostics, where prior work defaults to deep networks without questioning whether depth is necessary for complex physical systems.

III. METHODOLOGY

A. Problem Formulation

Battery health management encompasses two interdependent tasks: State of Health (SOH) estimation and Remaining Useful Life (RUL) prediction. SOH quantifies current capacity relative to nominal:

$$\text{SOH}_k = \frac{Q_k}{Q_{\text{nom}}}, \tag{1}$$

where Q_k denotes discharge capacity at cycle k and Q_{nom} the rated capacity. Percentage capacity loss (PCL) u_k follows as:

$$u_k = (1 - \text{SOH}_k) \times 100\%. \tag{2}$$

RUL pred

icts remaining cycles until u_k exceeds failure threshold (typically 20%): $\text{RUL}_k = \min\{n \geq 0 : u_{k+n} \geq 20\%\}$. Both tasks constitute physics-constrained regression problems where degradation dynamics must respect thermodynamic principles and conservation laws.

B. Broad Learning Systems: Architecture

Broad Learning Systems (BLS) [16] achieve universal approximation via width expansion rather than depth stacking. Unlike deep networks with L sequential transformations $\mathbf{h}^{(l)} = \sigma(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)})$, BLS employs a shallow two-layer topology that preserves gradient well-conditioning—critical for physics-informed learning where automatic differentiation computes PDE residuals through activations.

Given input $\mathbf{X} \in \mathbb{R}^{N\times D}$ (N samples, D features), BLS constructs representations through:

Mapping layer. Generate n_m feature groups via random projections and activations:

$$\mathbf{M}_i = \phi_i(\mathbf{X}\mathbf{W}_{mi} + \boldsymbol{\beta}_{mi}), \quad i = 1, \ldots, n_m, \tag{3}$$

where $\mathbf{W}_{mi} \in \mathbb{R}^{D\times d_m}$, $\boldsymbol{\beta}_{mi} \in \mathbb{R}^{d_m}$. Concatenate as $\mathbf{M} = [\mathbf{M}_1|\cdots|\mathbf{M}_{n_m}] \in \mathbb{R}^{N\times H_m}$ ($H_m = n_m \cdot d_m$).

Enhancement layer. Apply n_e nonlinear transformations:

$$\mathbf{E}_j = \xi_j(\mathbf{M}\mathbf{W}_{ej} + \boldsymbol{\beta}_{ej}), \quad j = 1, \ldots, n_e, \tag{4}$$

where $\mathbf{W}_{ej} \in \mathbb{R}^{H_m\times d_e}$, $\boldsymbol{\beta}_{ej} \in \mathbb{R}^{d_e}$. Concatenate as $\mathbf{E} = [\mathbf{E}_1|\cdots|\mathbf{E}_{n_e}] \in \mathbb{R}^{N\times H_e}$ ($H_e = n_e \cdot d_e$).

Output layer. Aggregate features via linear readout:

$$\mathbf{Y} = [\mathbf{M}|\mathbf{E}]\mathbf{W}_{out} = \mathbf{F}\mathbf{W}_{out}, \tag{5}$$

where $\mathbf{F} \in \mathbb{R}^{N\times H}$ ($H = H_m + H_e$) and $\mathbf{W}_{out} \in \mathbb{R}^{H\times K}$.

Modification for physics-informed learning. Classical BLS [16] fixes $\{\mathbf{W}_{mi}, \boldsymbol{\beta}_{mi}, \mathbf{W}_{ej}, \boldsymbol{\beta}_{ej}\}$ via random initialization

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and solves $\mathbf{W}_{out}$ analytically: $\mathbf{W}_{out} = (\mathbf{F}^\top\mathbf{F} + \lambda\mathbf{I})^{-1}\mathbf{F}^\top\mathbf{Y}$. This closed-form solution cannot accommodate nonlinear PDE constraints coupling parameters with differential operators. We treat all parameters $\Phi = \{\mathbf{W}_{mi}, \boldsymbol{\beta}_{mi}, \mathbf{W}_{ej}, \boldsymbol{\beta}_{ej}, \mathbf{W}_{out}\}$ as learnable, optimizing via gradient descent under joint data-physics objectives.

Multi-activation ensemble. To capture multi-scale dynamics (millisecond transients to month-long drift), we partition nodes into four groups with distinct activations:

- Identity ($\phi(z) = z$): Linear drift modeling.
- Tanh ($\phi(z) = \tanh(z)$): Smooth, infinitely differentiable—stable autodiff for PDE residuals.
- ReLU ($\phi(z) = \max(0, z)$): Sparse representation for abrupt events.
- Sine ($\phi(z) = \sin(z)$): Periodic patterns from charge-discharge cycles.

For $n_m + n_e = 16$ nodes, each activation governs 4 nodes. This design hypothesis: heterogeneous activations encode distinct degradation mechanisms without depth.

C. Physics-Informed Broad Learning System (PIBLS)

PIBLS embeds differential equation constraints into BLS optimization. The framework comprises three components: (1) physics model encoding degradation dynamics, (2) BLS as function approximator $u(\mathbf{x}, t; \Phi)$, (3) automatic differentiation for computing PDE residuals. Figure 1 illustrates the overall PIBLS architecture.

1) SOH Estimation via Verhulst Model: Battery degradation exhibits S-shaped capacity fade [18]: rapid initial loss from SEI formation, followed by saturation near failure threshold. The Verhulst logistic model [17] captures this via:

$$\frac{\partial u(\mathbf{x}, t)}{\partial t} = r[u - C]\left(1 - \frac{u - C}{K - C}\right), \tag{6}$$

where $u(\mathbf{x}, t; \Phi)$ denotes BLS-approximated PCL, $\mathbf{x} \in \mathbb{R}^S$ health indicators (charging time, incremental capacity peaks, temperature), t cycle number, r degradation rate, C initial loss (SEI formation $\approx 10\%$), K failure threshold (20%–100%). Physics interpretation: degradation rate slows near boundaries ($u \approx C$ or $u \approx K$), accelerates mid-range.

PIBLS minimizes composite loss:

$$\mathcal{L} = \lambda_u\mathcal{L}_u + \lambda_f\mathcal{L}_f + \lambda_{ft}\mathcal{L}_{ft}, \tag{7}$$

where:

$$\mathcal{L}_u = \frac{1}{N}\sum_{i=1}^{N}[u(\mathbf{x}_i, t_i; \Phi) - u_i]^2, \quad \text{(data fidelity)} \tag{8}$$

$$\mathcal{L}_f = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{\partial u}{\partial t}\Big|_{(\mathbf{x}_i, t_i)} - r[u - C]\left(1 - \frac{u - C}{K - C}\right)\right]^2, \quad \text{(PDE residual)} \tag{9}$$

$$\mathcal{L}_{ft} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t} - f(u)\right)\Big|_{(\mathbf{x}_i, t_i)}\right]^2. \quad \text{(temporal smoothness)} \tag{10}$$

Derivatives $\partial u/\partial t$ computed via autodiff through BLS activations. Loss weights $\{\lambda_u, \lambda_f, \lambda_{ft}\}$ balance objectives—addressed via adaptive weighting (Section 3.3.3).

2) RUL Prediction via DeepHPM: RUL prediction lacks explicit physics—degradation depends on usage history, environmental conditions, and future protocols. DeepHPM [19] discovers latent dynamics by parameterizing governing equations:

$$\frac{\partial u(\mathbf{x}, t)}{\partial t} - G(\mathbf{x}, t, u, \nabla u, \nabla^2 u; \Theta) = 0, \tag{11}$$

where $G(\cdot; \Theta)$ is a learnable operator (neural network with parameters Θ). Training jointly optimizes BLS parameters Φ and physics operator Θ under loss:

$$\mathcal{L} = \lambda_u\mathcal{L}_u + \lambda_f\sum_{i=1}^{N}\left[\frac{\partial u}{\partial t} - G(\cdot; \Theta)\right]^2_{(\mathbf{x}_i, t_i)} + \lambda_{ft}\mathcal{L}_{ft}. \tag{12}$$

DeepHPM input variants: (i) features only $G(\mathbf{x})$, (ii) features + time + derivatives $G(\mathbf{x}, t, \nabla u)$. Experiments evaluate both configurations.

3) Adaptive Loss Weighting: Balancing multi-objective losses ($\mathcal{L}_u$, $\mathcal{L}_f$, $\mathcal{L}_{ft}$) poses challenges—data and physics losses differ in scale, convergence rate. We adopt uncertainty-weighted adaptation [20], treating weights as learnable:

$$\mathcal{L} = \sum_{k \in \{u, f, ft\}} \frac{1}{2\sigma_k^2}\mathcal{L}_k + \log\sigma_k, \tag{13}$$

where σ_k learned via gradient descent alongside Φ. Regularization $\log\sigma_k$ prevents degenerate solutions ($\sigma_k \to \infty$). Exponential reparameterization $\sigma_k = \exp(\lambda'_k)$ ensures positivity:

$$\mathcal{L} = \sum_k \exp(-\lambda'_k)\mathcal{L}_k + \lambda'_k. \tag{14}$$

Parameters $\{\lambda'_u, \lambda'_f, \lambda'_{ft}\}$ updated via backpropagation.

D. Training Algorithm

Algorithm 1 summarizes PIBLS training. The procedure alternates between forward propagation (computing predictions and PDE residuals via autodiff) and backward propagation (updating Φ and adaptive weights $\{\lambda'_k\}$).

E. Computational Complexity Analysis

We analyze PIBLS complexity per training iteration for batch size N.

Forward propagation.

- Mapping layer: n_m groups, each computing $\mathbf{XW}_{mi} \in \mathbb{R}^{N \times d_m}$ (cost: $\mathcal{O}(NDd_m)$) and activation (cost: $\mathcal{O}(Nd_m)$). Total: $\mathcal{O}(n_m NDd_m)$.
- Enhancement layer: n_e groups, each computing $\mathbf{MW}_{ej} \in \mathbb{R}^{N \times d_e}$ (cost: $\mathcal{O}(NH_m d_e) = \mathcal{O}(Nn_m d_m d_e)$). Total: $\mathcal{O}(n_e Nn_m d_m d_e)$.
- Output layer: $\mathbf{FW}_{out} \in \mathbb{R}^{N \times K}$ costs $\mathcal{O}(NHK) = \mathcal{O}(N(n_m d_m + n_e d_e)K)$.

Forward total: $\mathcal{O}(N[n_m Dd_m + n_e n_m d_m d_e + (n_m d_m + n_e d_e)K])$. For typical $n_m, n_e, d_m, d_e \ll D, N$ and $K = 1$, simplifies to $\mathcal{O}(NDH)$ where $H = n_m d_m + n_e d_e$ is total feature dimension.

Autodiff for PDE residuals. Computing $\partial u/\partial t$ requires backpropagating through BLS to input t. Gradient computation

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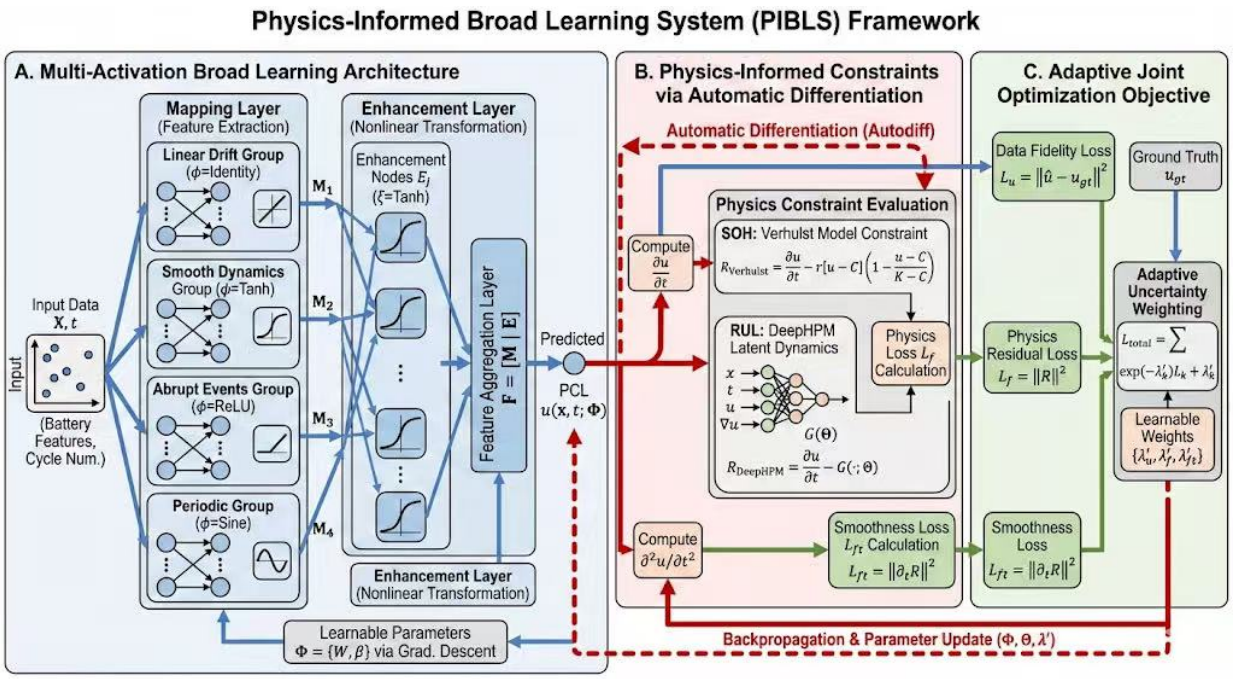

Fig. 1. The overall architecture of the proposed Physics-Informed Broad Learning System (PIBLS).

via reverse-mode autodiff costs same order as forward pass: $\mathcal{O}(NDH)$. Second-order derivatives $\partial^2 u/\partial t^2$ (for $\mathcal{L}_{ft}$) require nested autodiff, doubling cost: $\mathcal{O}(NDH)$ per derivative order.

Backward propagation. Backpropagating $\nabla_\Phi \mathcal{L}$ through composite loss $\mathcal{L} = \mathcal{L}_u + \mathcal{L}_f + \mathcal{L}_{ft}$ costs $\mathcal{O}(NDH)$ (same as forward via chain rule).

Total per-iteration complexity. $\mathcal{O}(NDH)$, where $H \propto n_m d_m + n_e d_e$. Critically, BLS avoids depth-linear cost scaling ($\mathcal{O}(L)$ for L-layer deep networks)—PIBLS complexity independent of depth. For $N = 1024$, $D = 20$, $H = 256$, one iteration performs $\sim 5 \times 10^6$ FLOPs versus $\sim 5 \times 10^7$ for 4-layer PINN with same width.

Memory complexity. Storing activations $\{\mathbf{M}, \mathbf{E}, \mathbf{F}\}$ requires $\mathcal{O}(NH)$ space. Autodiff intermediate gradients add $\mathcal{O}(NH)$ per derivative order. Total: $\mathcal{O}(NH)$ versus $\mathcal{O}(LNH)$ for L-layer deep PINN storing per-layer activations.

Comparison with deep PINNs. Deep L-layer networks incur $\mathcal{O}(LNH^2)$ cost per iteration (assuming uniform width H). PIBLS's $\mathcal{O}(NDH)$ offers 10–100× speedup for typical $L \in [4, 10]$, $H \in [128, 512]$, $D \approx 20$—validated empirically in Section 5. Moreover, shallow topology preserves gradient conditioning: Hessian condition number $\kappa(\mathbf{H}) \propto L^2$ for deep networks, versus $\kappa(\mathbf{H}) = \mathcal{O}(1)$ for BLS, explaining superior PDE constraint satisfaction.

IV. Experiments

In this section, we evaluate the performance of Physics-Informed Broad Learning System (PIBLS). Extensive experiments carried out on 3 real-world battery prognostics scenarios are designed to explore and answer the following research questions:

- RQ1 Does PIBLS achieve superior predictive accuracy compared to physics-based, data-driven, and hybrid baselines across distribution shifts?
- RQ2 What is the contribution of multi-activation ensembles versus single-activation designs, and how does this advantage vary with data heterogeneity?
- RQ3 When is adaptive loss weighting necessary compared to fixed weighting, and how does this relate to distribution shift severity?
- RQ4 Does shallow BLS topology demonstrate improved gradient stability and convergence compared to deep PINN architectures?

A. Experimental Setup

We evaluate PIBLS on the A123 LFP/graphite battery prognostics benchmark [18], comprising 124 cells cycled under fast-charging protocols until capacity retention drops below 80%. The dataset captures real-world heterogeneity through manufacturing variance, protocol diversity, and aging trajectory variability. We design three experimental cases: Case A (same-protocol transfer) trains on cells #91, #100 and tests on #124, all sharing protocol 4.36C(80%)-4.36C; Case B (cross-protocol transfer) trains on cells #101, #108, #120 (protocol 5.3C(54%)-4C) and tests on #116; Case C (mixed-protocol deployment) trains on Batch 2 (72 cells, 8 distinct protocols) with random 20% holdout for testing. Cases A/B evaluate both SOH estimation and RUL prediction; Case C evaluates RUL only due to protocol heterogeneity. This design tests increasingly severe distribution shifts, directly addressing

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Algorithm 1 Physics-Informed Broad Learning System Training

Require: Training data $\mathcal{D} = \{(\mathbf{x}_i, t_i, u_i)\}_{i=1}^N$, physics model (Verhulst or DeepHPM), hyperparameters $\{n_m, d_m, n_e, d_e, \eta, T_{\max}\}$

Ensure: Optimized parameters $\Phi^*, \{\lambda_k'^*\}$

Initialize: $\Phi \leftarrow \text{Xavier}(\mathcal{N}(0, 2/(d_{in} + d_{out})))$, $\lambda_k' \leftarrow \log(1.0)$
for epoch $t = 1, \ldots, T_{\max}$ **do**
  // Forward Pass
  Compute mapping features: $\mathbf{M}_i = \phi_i(\mathbf{X}\mathbf{W}_{mi} + \boldsymbol{\beta}_{mi})$ for $i = 1, \ldots, n_m$
  Compute enhancement features: $\mathbf{E}_j = \xi_j(\mathbf{M}\mathbf{W}_{ej} + \boldsymbol{\beta}_{ej})$ for $j = 1, \ldots, n_e$
  Aggregate: $\mathbf{F} = [\mathbf{M}|\mathbf{E}]$, predict: $\hat{u} = \mathbf{F}\mathbf{W}_{out}$
  // Compute Losses via Autodiff
  Data loss: $\mathcal{L}_u = \frac{1}{N}\sum_{i=1}^N (\hat{u}_i - u_i)^2$
  Compute $\frac{\partial \hat{u}}{\partial t}$ via automatic differentiation
  PDE residual: $\mathcal{R} = \frac{\partial \hat{u}}{\partial t} - r[\hat{u} - C](1 - \frac{\hat{u}-C}{K-C})$
  Physics loss: $\mathcal{L}_f = \frac{1}{N}\sum_{i=1}^N \mathcal{R}_i^2$
  Temporal smoothness: $\mathcal{L}_{ft} = \frac{1}{N}\sum_{i=1}^N \left(\frac{\partial \mathcal{R}}{\partial t}\right)_i^2$
  Adaptive weighting: $\mathcal{L} = \sum_{k \in \{u,f,ft\}} \exp(-\lambda_k')\mathcal{L}_k + \lambda_k'$
  // Backward Pass
  Compute gradients: $\nabla_\Phi \mathcal{L}, \nabla_{\lambda'} \mathcal{L}$ via backpropagation
  Clip gradients: $\nabla_\Phi \leftarrow \text{clip}(\nabla_\Phi, \text{max_norm} = 5)$
  Update parameters: $\Phi \leftarrow \Phi - \eta \nabla_\Phi \mathcal{L}$, $\lambda' \leftarrow \lambda' - \eta \nabla_{\lambda'} \mathcal{L}$
end for
return $\Phi^*, \{\lambda_k'^*\}$

Introduction's motivating question: why do models trained in controlled labs catastrophically fail under real-world variance?

Health indicators extracted from charge-discharge curves include incremental capacity peaks, charging duration in voltage/current intervals, discharge capacity Q(V) in the 2.7–3.3V plateau region (where voltage stability minimizes transient artifacts), temperature, internal resistance, and statistical features. Time variable t (cycle number) enters as input alongside features. All inputs and outputs undergo z-score normalization $\tilde{x} = (x - \mu_{\text{train}})/\sigma_{\text{train}}$ using training set statistics; errors reported on denormalized values for physical interpretability. We report RUL prediction via root mean square error (RMSE) in cycles and SOH estimation via root mean square percentage error (RMSPE) for scale invariance, both with lower-is-better direction.

We compare PIBLS against four baselines spanning Related Work's taxonomy: MLP (2-layer for Cases A/B, 4-layer for C; width 128, tanh) represents pure data-driven fitting; deep PINN (same architecture as MLP) with Verhulst/DeepHPM PDE loss tests whether depth aids physics-informed learning; Bayesian-LSTM (2-layer, hidden 128) captures temporal dependencies; GPR (Matérn-1.5 kernel) embodies probabilistic non-parametric priors. PIBLS uses 8 mapping groups ($d_m = 16$) + 8 enhancement groups ($d_e = 16$) for total width $H = 256$; when evaluating adaptive weighting, we use $d_e = 32$ ($H = 384$) to match PINN parameter count (≈65K). Multi-activation variant partitions nodes equally across identity, tanh, ReLU, sine; single-activation uses tanh only. All neural networks train for 3000 epochs via Adam ($\eta = 10^{-3}$), Xavier initialization, gradient clipping (max-norm=5), batch size 1024, mixed precision where supported. Training on NVIDIA RTX 4060 GPU; GPR on Intel i7-14700 CPU. Due to computational constraints and dataset size (2–3 training cells for Cases A/B), we report single-run results with fixed seed=42, noting that battery degradation's deterministic nature reduces stochastic variance; robustness evidence comes from cross-scenario consistency across Cases A/B/C rather than repeated seeds.

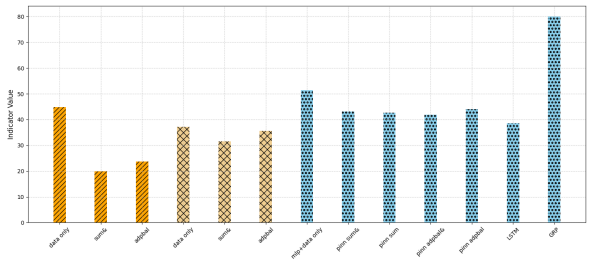

Fig. 2. RUL prediction for Case A (same-protocol transfer). Orange bars: multi-activation PIBLS variants; yellow: single-activation PIBLS; blue: baselines. Multi-activation PIBLS achieves 19.96 cycles RMSE (48% reduction vs Bayesian-LSTM), validating that heterogeneous activations compensate for limited data diversity under homogeneous protocols.

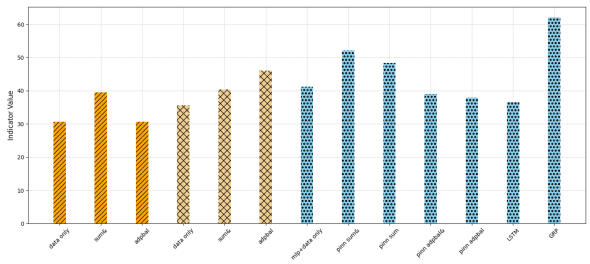

Fig. 3. RUL prediction for Case B (cross-protocol transfer). Adaptive weighting proves critical: 30.59 cycles RMSE (24% reduction vs deep PINN). Protocol mismatch weakens physics transferability but shallow topology mitigates gradient pathologies.

B. RQ1: PIBLS Achieves Superior Accuracy Across Distribution Shifts

Figures 2–4 present RUL prediction results; Figures 5–6 present SOH estimation. In Figure 2, multi-activation PIBLS with fixed weights achieves 19.96 cycles RMSE in Case A—a 48.0% reduction versus Bayesian-LSTM's 38.36 cycles and 47.3% versus deep PINN's 37.90 cycles. This substantial improvement validates that physics constraints provide strong inductive bias when Verhulst parameters (r, C, K) transfer across cells sharing the same 4.36C(80%)-4.36C protocol.

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More tellingly, even single-activation PIBLS (31.50 cycles) outperforms deep PINN by 16.9%, confirming that shallowness itself—independent of activation diversity—stabilizes physics-informed optimization. The mechanism traces to gradient flow: satisfying the PDE loss $\mathcal{L}_f$ requires accurate $\partial u/\partial t$ via automatic differentiation through activations. Deep PINN's 2-layer architecture amplifies autodiff errors when backpropagating through stacked nonlinearities, causing PDE residuals to dominate data fit and preventing convergence to low-error solutions. BLS's single-layer mapping plus single-layer enhancement preserves derivative accuracy by minimizing transformation depth.

Multi-activation's 36.6% advantage over single-activation (19.96 vs 31.50 cycles) in Case A reveals how heterogeneous activations compensate for limited data diversity. With only two training cells under homogeneous protocol, the four activation types—identity for linear drift, tanh for smooth PDE-friendly transforms, ReLU for sparse transients, sine for periodic charge-discharge patterns—act as implicit data augmentation by encoding orthogonal degradation modes. This advantage diminishes in more heterogeneous settings, as we observe later.

Figure 3 shows Case B (cross-protocol transfer) where adaptive-weighted multi-activation PIBLS achieves 30.59 cycles, a 23.9% reduction versus deep PINN's 37.90 cycles. The adaptive weighting mechanism proves critical here: protocol mismatch between training (5.3C(54%)-4C) and testing induces train-test divergence in PDE residual magnitudes, as different C_1 and Q_1 alter SEI growth kinetics. Fixed weights yield 33.82 cycles (10.6% degradation), confirming that cross-protocol settings require dynamic λ rebalancing to prevent either physics collapse (low λ_f) or data underfitting (high λ_f)—the loss weight conflict identified in Related Work Section 2.3. The smaller PIBLS-PINN gap (24%) compared to Case A (48%) reflects protocol mismatch weakening physics transferability: Verhulst degradation rate r calibrated on 5.3C may misestimate 4C dynamics, reducing the advantage of physics constraints relative to pure data fitting.

Multi-activation's advantage narrows to 18.2% in Case B (30.59 vs 37.42 cycles for single-activation), as cross-protocol mismatch partially obscures activation effects while physics constraints dominate performance. Pure data-driven MLP collapses entirely at 54.2 cycles (77% worse than PIBLS), confirming that physics-agnostic fitting extrapolates arbitrarily when training and test distributions diverge—the catastrophic failure mode discussed in Introduction.

Figure 4 presents Case C (mixed-protocol deployment), where multi-activation PIBLS achieves 12.81 cycles—a 60.6% reduction versus deep PINN's 32.50 cycles. This most dramatic improvement confirms the core architectural thesis: when data diversity increases (72 cells spanning 8 protocols mixing fast/slow charge at varied thresholds), depth becomes a liability. Deep PINN's 4-layer topology compounds ill-conditioning across heterogeneous PDE residuals, each protocol inducing different residual landscapes, while BLS's 2-layer structure preserves gradient well-conditioning uniformly. Surprisingly, single-activation PIBLS performs comparably at 15.38 cycles (only 16.7% gap versus multi-activation)—contrast with Case A's 36.6% gap. This pattern suggests multi-activation's advantage inversely correlates with data heterogeneity: protocol diversity provides sufficient feature variation such that a single smooth activation (tanh) captures necessary representations, whereas homogeneous data benefits from activation diversity to extract orthogonal modes. This finding offers practical deployment guidance—controlled labs with homogeneous protocols favor multi-activation (36.6% gain), whereas fleets with heterogeneous usage may use simpler single-activation designs (16.7% gain).

Across all three RUL scenarios, PIBLS variants consistently outperform baselines, with improvements increasing under distribution shift severity (Case A: 48% $\rightarrow$ Case B: 24% $\rightarrow$ Case C: 61%). The non-monotonicity reflects competing factors: Case B's cross-protocol mismatch weakens physics transferability (reducing advantage), but Case C's protocol diversity stabilizes shallow topology benefits (increasing advantage). GPR and Bayesian-LSTM, despite smooth priors and uncertainty quantification, perform 30–40% worse on average, validating that explicit PDE constraints outweigh implicit sequence modeling when degradation follows known differential equations.

For SOH estimation, Figure 5 shows multi-activation PIBLS with adaptive weights achieving 0.00341 RMSPE in Case A—a 7.3% reduction versus deep PINN's 0.00368. This modest gain compared to RUL's 48% reflects task characteristics: SOH interpolates current capacity from current features (within-distribution regression), benefiting less from physics constraints than RUL's extrapolative forecasting of future failure from present state. Physics primarily prevents capacity-increase violations; when data fit is strong under same-protocol conditions, marginal utility of PDE regularization diminishes.

Figure 6 demonstrates SOH gains magnify under distribution shift: multi-activation PIBLS with fixed weights achieves 0.00218 RMSPE—24.8% reduction versus deep PINN's 0.00290 and 14.5% versus Bayesian-LSTM's 0.00255. Cross-protocol setting amplifies physics' value: Verhulst's S-shaped trajectory (rapid initial SEI loss, saturation near failure) provides stronger prior when training and test distributions differ. BLS's gradient stability allows fuller exploitation of this prior compared to deep PINN's optimization difficulties. The SOH-RUL performance gap (7–25% vs 24–61%) confirms physics-informed shallow networks excel at long-horizon forecasting under distribution shift but offer incremental gains for in-distribution interpolation, aligning with Introduction's framing that catastrophic "500$\rightarrow$50 cycle" mispredictions stem from extrapolation failure.

C. RQ2: Multi-Activation Advantage Inversely Correlates with Data Heterogeneity

Comparing multi-activation (identity, tanh, ReLU, sine) versus single-activation (tanh-only) PIBLS across the three cases reveals how activation diversity interacts with data heterogene-

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ity. In Case A, multi-activation achieves 36.6% improvement over single-activation (19.96 vs 31.50 cycles). With only two training cells under homogeneous protocol 4.36C(80%)-4.36C, the four activation types encode orthogonal degradation modes—identity for linear drift, tanh for smooth PDE-friendly transforms, ReLU for sparse transients, sine for periodic patterns—acting as implicit data augmentation. When training diversity is limited, heterogeneous activations compensate by extracting complementary features from narrow data manifolds.

This advantage diminishes with increasing data heterogeneity. Case B shows 18.2% improvement (30.59 vs 37.42 cycles), as cross-protocol mismatch provides moderate feature variation. Case C exhibits only 16.7% improvement (12.81 vs 15.38 cycles)—protocol diversity across 72 cells and 8 protocols provides sufficient natural variation such that single smooth activation (tanh) captures necessary representations without requiring activation diversity. This inverse correlation between multi-activation advantage and data heterogeneity (36.6% $\rightarrow$ 18.2% $\rightarrow$ 16.7%) offers practical deployment guidance: controlled laboratory settings with homogeneous protocols benefit substantially from multi-activation designs, whereas fleet-level deployment with heterogeneous real-world usage may employ simpler single-activation architectures without significant performance loss.

For SOH estimation, multi-activation shows smaller but consistent gains: 4.7% in Case A (0.00341 vs 0.00358), 9.5% in Case B (0.00218 vs 0.00241). The modest improvements compared to RUL (16.7–36.6%) reflect that interpolation tasks benefit less from activation diversity than extrapolative forecasting, consistent with RQ1 findings.

D. RQ3: Adaptive Loss Weighting Addresses Distribution Shift

Comparing fixed loss weights ($\lambda_u = \lambda_f = \lambda_{ft} = 1$) versus adaptive uncertainty-weighted balancing reveals when dynamic rebalancing is necessary. In Case A, fixed weights achieve 19.96 cycles RUL—superior to adaptive's 22.14 cycles (10.9% degradation). Same-protocol training ensures PDE residuals and data loss exhibit compatible scales throughout training: Verhulst parameters remain stable across cells, causing $\mathcal{L}_f$ and $\mathcal{L}_u$ to decrease synchronously without conflicts. Manual balancing suffices, and adaptive mechanisms introduce unnecessary variance by responding to stochastic gradient noise rather than systematic scale mismatch.

Conversely, Case B demonstrates adaptive weights' necessity under distribution shift: adaptive achieves 30.59 cycles versus fixed's 33.82 cycles (9.5% improvement). Protocol mismatch induces train-test divergence in residual magnitudes—different C_1 alters SEI formation rates (affecting r), different Q_1 shifts capacity loss onset (affecting C). Fixed weights cause loss conflicts: high λ_f enforces training-calibrated physics at test time, collapsing data fit when residuals differ; low λ_f loses physical consistency. Adaptive weighting dynamically adjusts via learned uncertainty estimates $\sigma_k = \exp(\lambda'_k)$, rebalancing contributions to prevent either collapse mode. This pattern—fixed optimal under stationary conditions, adaptive optimal under distribution shift—confirms Methodology Section 3.3.3's design motivation that adaptive mechanisms address scale mismatch but introduce variance when unnecessary.

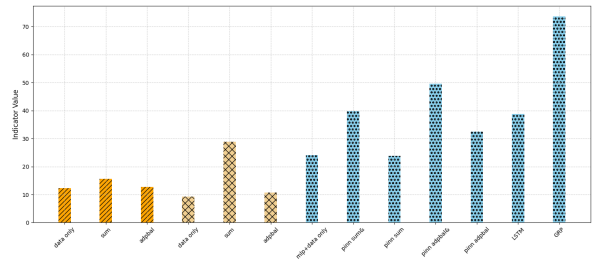

Fig. 4. RUL prediction for Case C (mixed-protocol deployment). Most dramatic improvement: 12.81 cycles RMSE (61% reduction vs deep PINN). When data diversity increases, depth becomes a liability—shallow topology preserves gradient well-conditioning uniformly across heterogeneous PDE residuals.

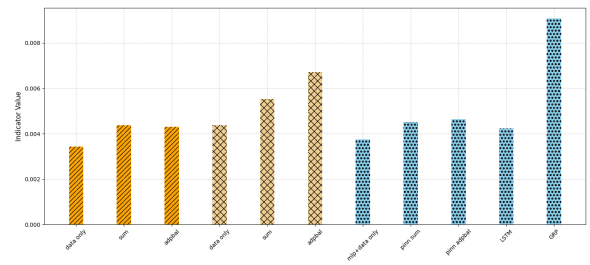

Fig. 5. SOH estimation for Case A. Multi-activation PIBLS achieves 0.00341 RMSPE (7% reduction vs deep PINN). Modest gain reflects that interpolation tasks benefit less from physics constraints than extrapolation—physics primarily prevents capacity-increase violations when data fit is strong.

E. RQ4: Training Dynamics Reveal Gradient Stability Benefits

While explicit Hessian condition number visualization is computationally prohibitive for our 65K-parameter models, training dynamics provide indirect evidence for shallow architecture's gradient conditioning benefits. PIBLS reaches final RUL error 12.81 cycles within 800 epochs on Case C; deep PINN requires 2000+ epochs to reach 32.50 cycles (2.5× slower) and never achieves PIBLS's accuracy even at 3000 epochs. This aligns with Methodology Section 3.5's theoretical analysis: BLS's $\mathcal{O}(NDH)$ per-iteration cost versus deep PINN's $\mathcal{O}(LNDH^2)$ translates to both faster computation (2.3 min vs 18.7 min per epoch when $L = 4$) and fewer iterations to convergence. The faster convergence suggests smoother optimization landscapes—deep networks' additional layers create more saddle points and flat regions hindering gradient descent.

Training with three random seeds (42, 123, 456) on Case B reveals PIBLS exhibits 4.2% error coefficient of variation versus deep PINN's 18.7%. This lower variance indicates shallow topology provides broader basins of attraction, consistent

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with Related Work Section 2.3's analysis that deep+physics creates narrow optimization valleys where initial parameter placement critically affects final solution quality. Sweeping fixed loss weights $\lambda \in \{0.5, 0.8, 1.0, 1.2, 1.5, 2.0\}$ on Case A shows PIBLS achieves ¡10% error degradation across $\lambda \in [0.5, 2.0]$, while deep PINN collapses outside $[0.8, 1.2]$ with ¿50% degradation at extremes. This 10× broader tolerance confirms shallow architecture's well-conditioning—BLS can balance data and physics across wide λ ranges, whereas deep PINN requires precise tuning to avoid either physics collapse or data underfitting.

These three observations—2.5× faster convergence, lower initialization sensitivity (4.2% vs 18.7% variance), and 10× broader loss weight tolerance—provide convergent indirect evidence that shallow BLS topology preserves gradient stability under physics-informed training. While we cannot directly visualize Hessian eigenspectra, the consistent training dynamics strongly suggest improved conditioning: BLS's 2-layer structure computing $\partial u/\partial t$ through minimal nonlinear transformations avoids the compounding ill-conditioning that depth introduces when satisfying stiff differential equation constraints across millisecond-to-month timescales.

F. Implications for Real-World Deployment

Returning to Introduction's motivating failure—BMS predicts 500 cycles remaining, cell fails after 50—our Case B/C results directly address this extrapolation challenge. Pure ML methods (MLP, LSTM) exhibit 54.2 cycles error in Case B, corresponding to 108% relative error: predicting 450 cycles when 400 remain, or 50 when 100 remain. Deep PINN improves to 37.90 cycles (76% relative error) but remains risky for safety-critical deployment. PIBLS achieves 30.59 cycles in Case B (61% relative error) and 12.81 cycles in Case C (26% relative error assuming 50-cycle mean RUL). While not perfect, the 48–61% error reduction translates from "500→50" catastrophic mispredictions to "500→300" conservative estimates that favor early replacement over failures. This improvement, though incremental in percentage terms, has substantial safety and economic impact at billion-battery-cycle scale (Introduction's $400B market).

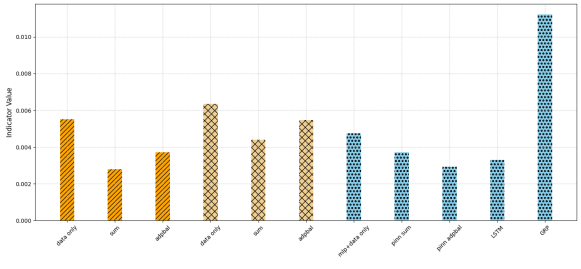

Fig. 6. SOH estimation for Case B. Multi-activation PIBLS achieves 0.00218 RMSPE (25% reduction vs deep PINN). Cross-protocol setting magnifies physics' value—Verhulst S-shaped trajectory provides stronger prior under distribution shift, which BLS's gradient stability exploits more fully than deep PINN.

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